

10、11 章作业答案

10.5 解. 设 $A(x_A, y_A)$ $C(x_C, y_C)$

则有 $x_A = l$ $y_A = l \tan \varphi$ $x_C = a \cos \varphi$ $y_C = a \sin \varphi$, 令 φ 为广义坐标

则由虚位移原理可知:

$$\begin{aligned} \delta W &= -Q \cos \varphi \cdot \frac{\partial y_C}{\partial \varphi} + Q \sin \varphi \frac{\partial x_C}{\partial \varphi} + P \frac{\partial y_A}{\partial \varphi} \\ &= -Q \cos \varphi \cdot a \cos \varphi - Q \sin \varphi a \sin \varphi + P \cdot l \frac{1}{\cos^2 \varphi} \\ &= -Q \cdot a + \frac{Pl}{\cos^2 \varphi} = 0 \end{aligned}$$

$$\Rightarrow P = \frac{Qa \cos^2 \varphi}{l}$$

10.8 解. 解除弹簧约束, 代之以 F_T , 如右图所示

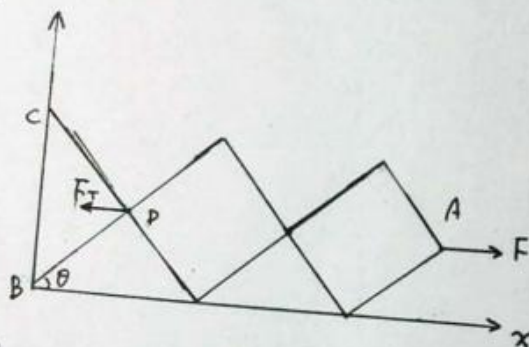
设 θ 为广义坐标, 平衡时, 由虚功原理可知

$$\delta W_F = F \cdot \delta x_A - F_T \delta x_D = 0 \quad (1)$$

$$\because x_A = 5x_D \quad \text{则} \quad \delta x_A = 5\delta x_D$$

$$\text{由条件知 } x_D = 0.5 \cos \theta \Rightarrow \delta x_D = -0.5 \sin \theta \delta \theta$$

$$\text{则} \quad \text{由} (1) \text{ 式可知 } \cos \theta = \frac{5F}{0.5k} = 0.25 \Rightarrow \theta = 75.5^\circ$$



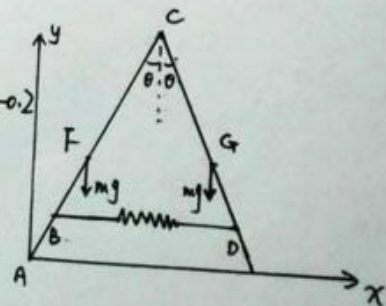
10.9 解. 如图所示, 建立平面直角坐标系, 取 θ 为广义坐标

$$\text{则} \quad y_F = 0.3 \cos \theta = y_G$$

$$\text{弹簧现长 } l = x_D - x_B = 1.2 \sin \theta \quad \text{弹簧伸长量 } \Delta l = l - l_0 = 1.2 \sin \theta - 0.2$$

$$\begin{aligned} \text{系统势能 } U &= 2mgh + \frac{1}{2}k\Delta l^2 \\ &= 34.3 \cos \theta + \frac{1}{2}k(1.2 \sin \theta - 0.2)^2 \end{aligned}$$

$$\frac{dU}{d\theta} = -34.3 \sin \theta + k(1.2 \sin \theta - 0.2) \times 1.2 \cos \theta = 0 \Rightarrow \theta = 10.3^\circ$$



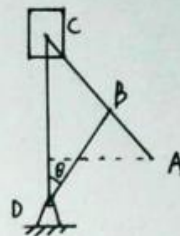
10.13. 解. 如右图所示, 以水平面为基准面, 可知整个系统势能

$$U = mg \left[a \cos \theta - \sqrt{c^2 - (a \sin \theta)^2} \cdot \frac{b}{c} \right] \quad (\text{取 } \theta \text{ 为广义坐标})$$

由势能原理可知.

$$\frac{dU}{d\theta} = \left[-a \sin \theta - \frac{b}{c} \cdot \frac{-2a^2 \sin \theta \cos \theta}{2\sqrt{c^2 - (a \sin \theta)^2}} \right] mg = 0$$

$$\Rightarrow \frac{b}{c} \cdot \frac{a \cos \theta}{\sqrt{c^2 - a^2 \sin^2 \theta}} = 1 \quad \Rightarrow \theta = 33^\circ$$



10.17. 解. 如右图所示, 以水平面为基准面, 可知整个系统势能

$$U = G_1 + G_2 + G_3$$

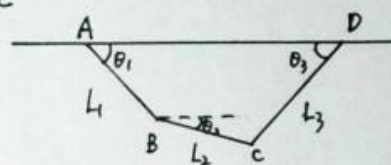
$$= -\frac{1}{2} mg L_1 \sin \theta_1 - mg (L_1 \sin \theta_1 + \frac{1}{2} L_2 \sin \theta_2) - \frac{1}{2} mg L_3 \sin \theta_3$$

$$\text{又} \because \sin \theta_2 = \frac{L_3 \sin \theta_3 - L_2 \sin \theta_1}{L_2}$$

$$\text{则令 } \theta_1, \theta_3 \text{ 为广义坐标, } U = -mg (L_1 \sin \theta_1 + L_3 \sin \theta_3)$$

$$\text{由势能原理可知 } \frac{dU}{d\theta_1} = 0, \quad \frac{dU}{d\theta_3} = 0$$

$$\Rightarrow \theta_1 = \theta_3 = 90^\circ \quad \theta_2 = \arcsin \frac{|L_3 - L_1|}{L_2}$$



10.25. 解. 建立如右图所示平面直角坐标系

$$\text{设 } A(x_A, y_A) \quad C(x_C, y_C)$$

由条件知: 细杆能随遇静止, 则由虚功原理可知

$$\delta W = 0 \Rightarrow \delta y_C = 0 \Rightarrow y_C = l \quad (\text{杆靠在铅垂墙上时})$$

$$\text{又} \because y_C = y_A + l \cos \varphi$$

$$\Rightarrow \begin{cases} y_A = l - l \cos \varphi \\ x_A = 2l \sin \varphi \end{cases}$$

$$\Rightarrow \frac{x_A^2}{4l^2} + \frac{(l - y_A)^2}{l^2} = 1$$

